Predicting Future Bike Accident Hotspots in Amsterdam

Using HDBSCAN Clustering & Multi-Source Data Analysis

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Data-Driven Approach

PREDICTION MODEL

Introduction to Bike Culture in Amsterdam

Mapping Accident Hotspots

Key Findings Future Enhancements

Introduction to Bike Culture in Amsterdam

Cycling in the Netherlands: More Than Transport

Cycling is a key part of Dutch culture.

- About 27% of trips are by bike.
- Healthy lifestyle and environment friendly.
- Amsterdam has over 500 km of bike paths.
- City design focuses on safety and easy access.
- Great cycling infrastructure.

Urban Cycling Safety: Ongoing Challenges

- **Intersections** remain key accident hotspots
- Rising cycling traffic increases potential risks
- Need for enhanced safety measures & smarter infrastructure
- Focus on data-driven urban planning for future improvements

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Data-Driven Approach

Introduction to Open Datasets

29,763 bike accidents analyzed (2014–2023)
- **Goal**: Predict future accident hotspots
- **Method**: HDBSCAN clustering + weather + injury data
- **Output**: 2025 risk predictions for Amsterdam

Key Statistics

- **29,763** total accidents
- **170** hotspot clusters identified
- **1,461** injury accidents
- **34** fatal accidents
- **95%** accident increase (2014-2023, 10years)

Impact: Every prediction saves lives

Data Sources and Collection Methods

Three Independent Datasets:

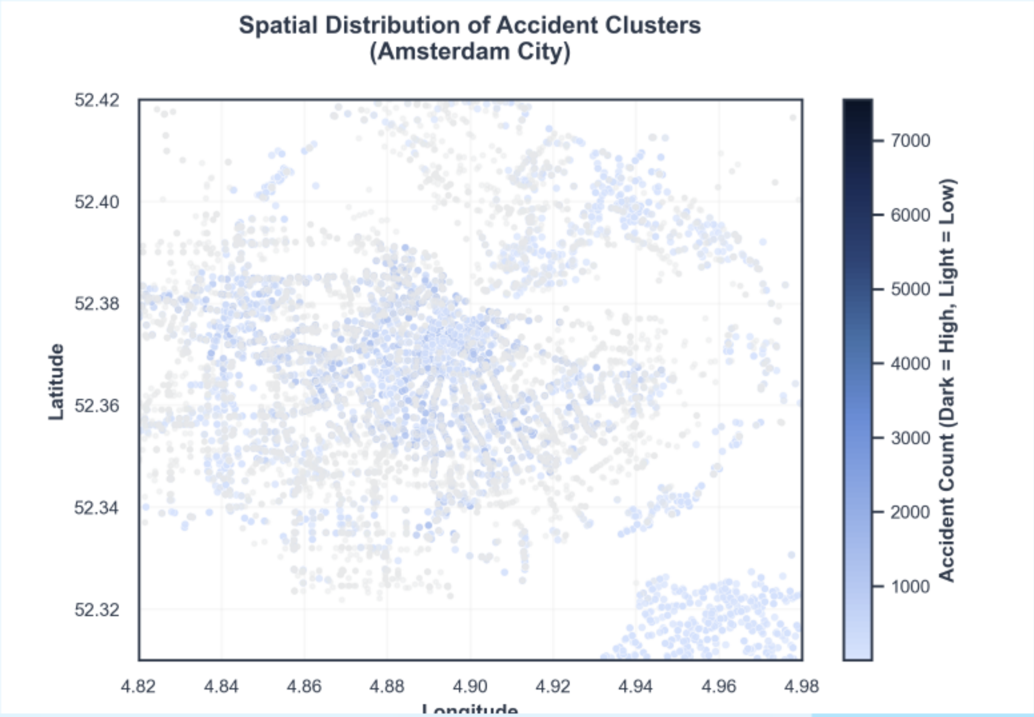
- **Coordinates:** 29,765 accident records with GPS
- **Weather**: 7 years temperature, precipitation, wind
- **Injury Data**: Severity + weekday patterns (1,803 locations)

Key: No data merging - preserved data integrity

METHODOLOGY

- **01** **Clean** coordinates   Amsterdam area filter
- **02** **Cluster** with HDBSCAN   170 hotspots found
- **03** **Integrate** weather risk factors
- **04** **Analyze** injury severity patterns
- **05** **Predict** 2025 hotspot risks

Quality: Silhouette Score 0.486



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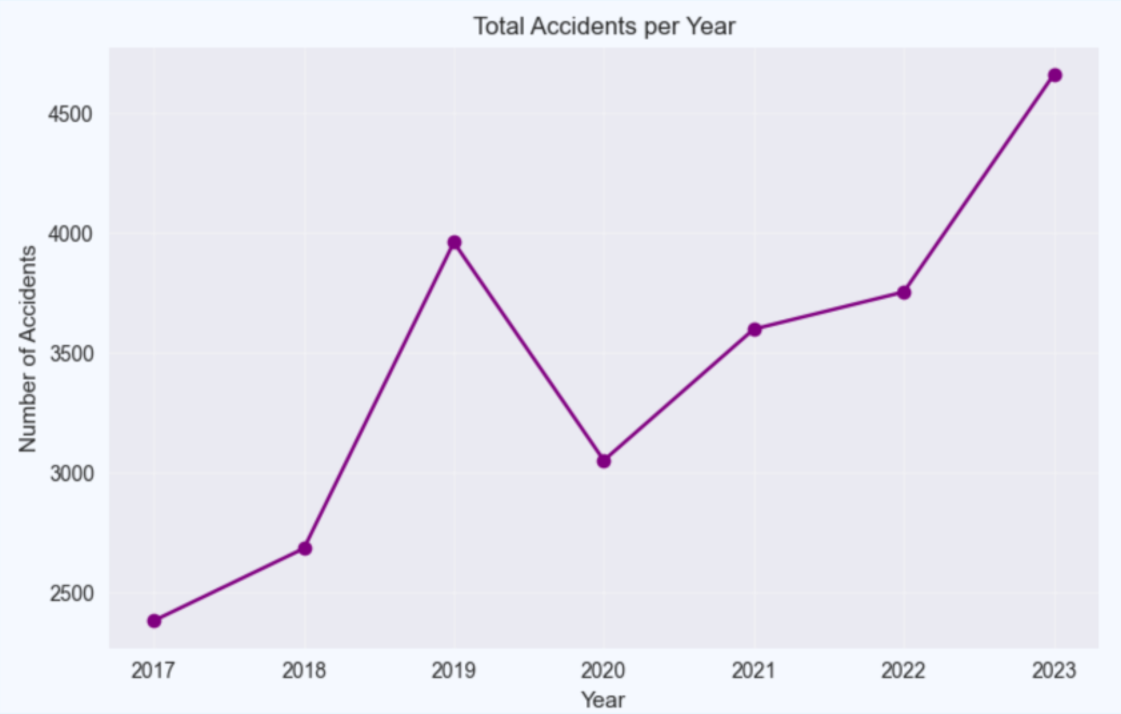

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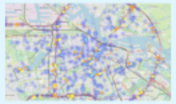
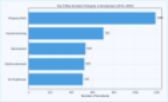
Mapping Accident Hotspots

CLUSTERING RESULTS

Performance Metrics:

- 170 hotspot clusters identified
- 9,986 noise points (isolated incidents)
- Largest hotspot: 1,189 accidents (Ringweg-West)
- Average per cluster: 116.3 accidents

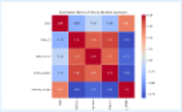


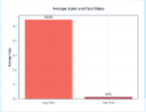


WEATHER IMPACT

Accident Growth:
2017: 2,384 accidents
2023: 4,660 accidents
Growth: 95% increase

Temperature: 10.6°C - 12.0°C
Precipitation: 1.5mm - 3.0mm average
Wind: 4.8m/s - 5.2m/s
Visibility: 22,500m - 23,400m





INJURY SEVERITY

Severity Metrics:

- 54.4% average injury rate
- 1.57% fatal accident rate

- Fatal × 3 + Injury × 2 + Damage × 1

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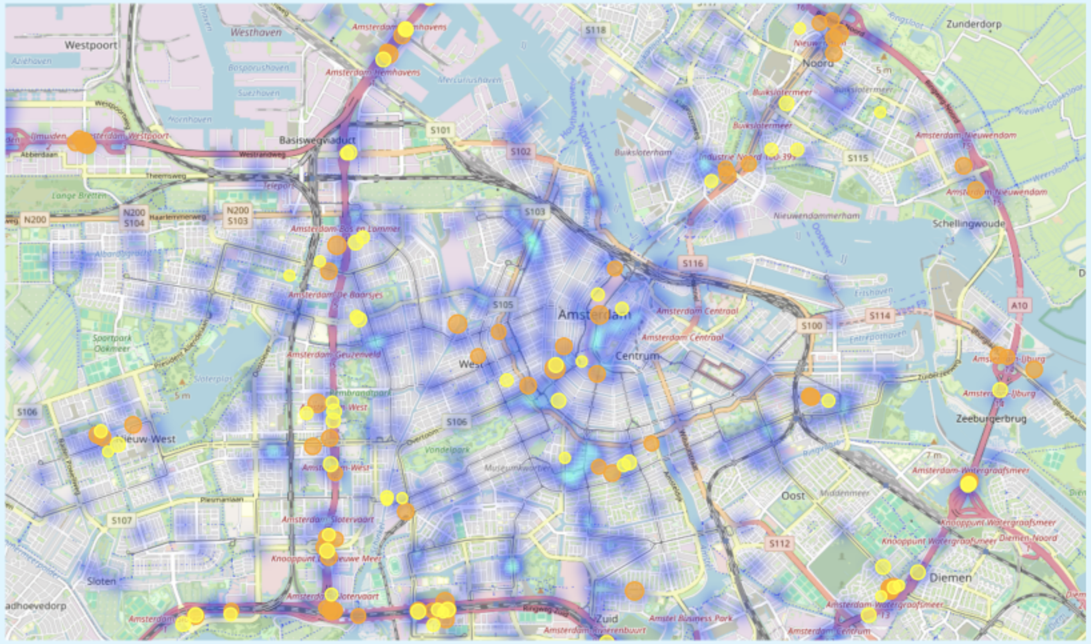

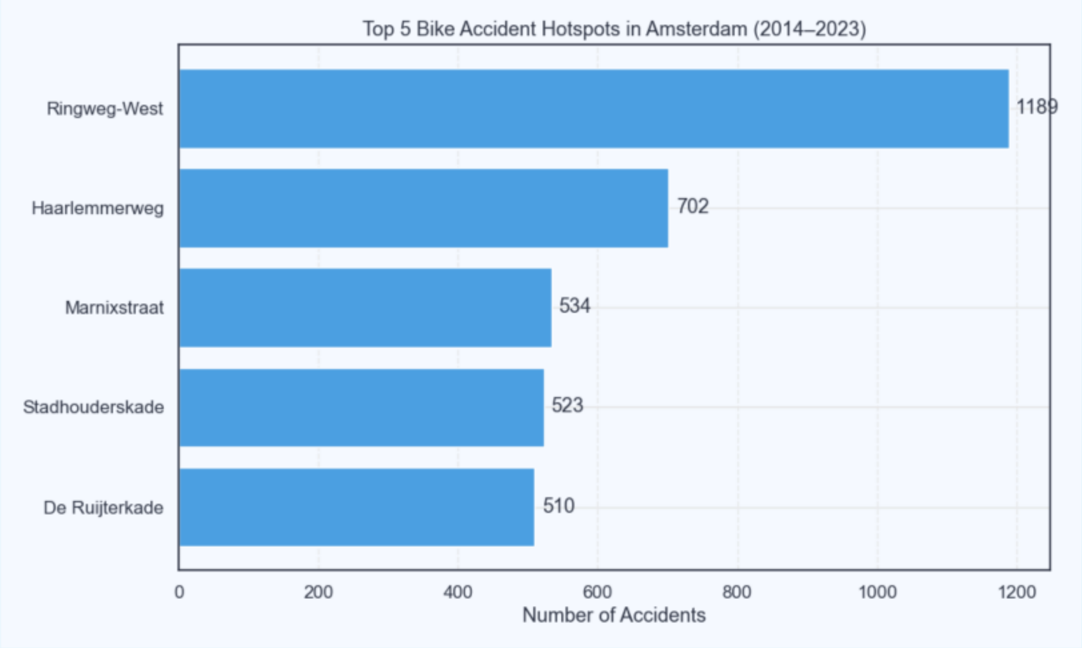

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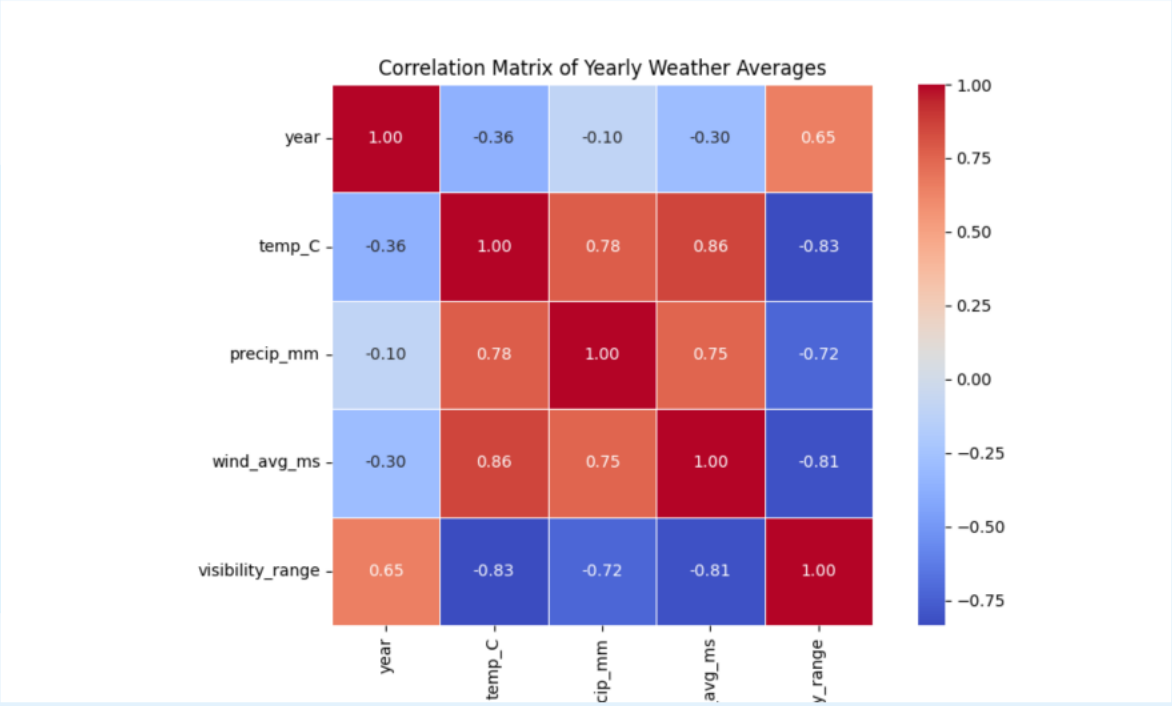

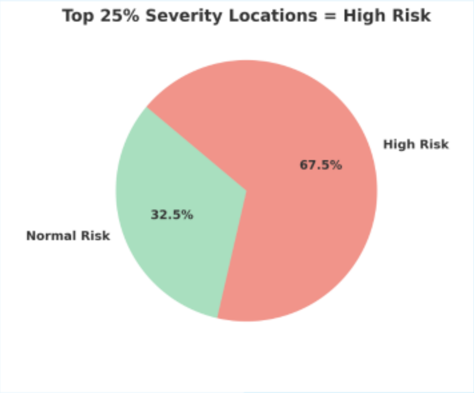

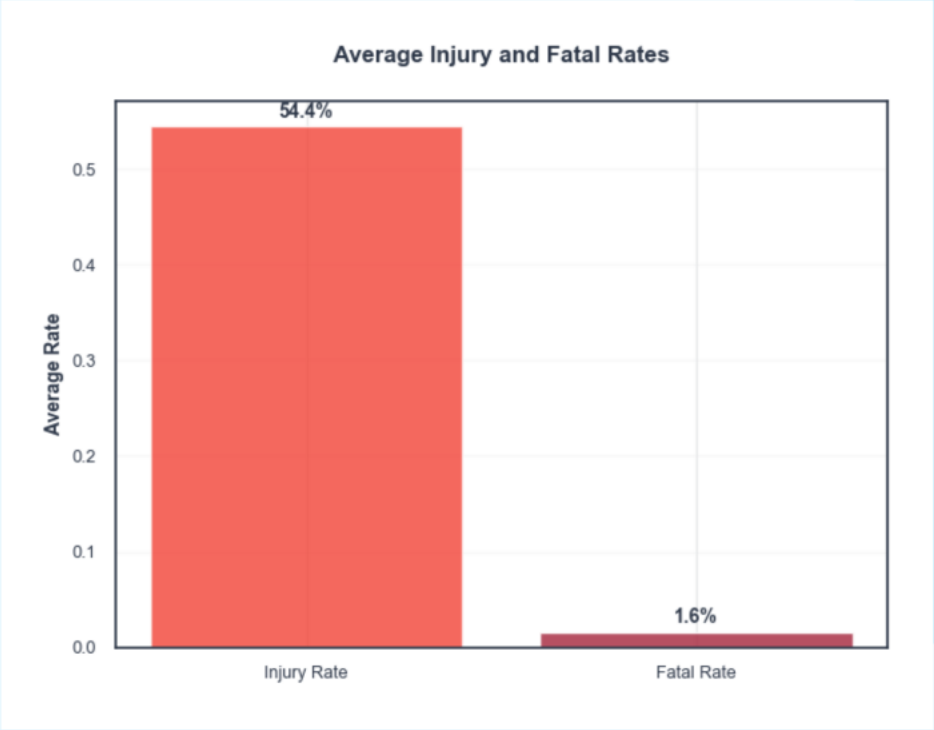

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PREDICTION MODEL

Scoring Formula (Weighted):

- Base Risk (40%): Historical accident frequency
- Weather Risk (30%): Temperature, precipitation, wind
- Temporal Weight (20%): Recent years emphasis
- Injury Risk (10%): High-severity proximity

Risk Levels:
High: Score > 0.7
Medium: Score 0.4-0.7
Low: Score < 0.4

2025 PREDICTIONS -NORMAL

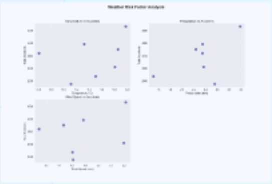
Top Predicted Hotspots:
1. **Cluster 117** - Ringweg-West (0.555)
2. **Cluster 1**- Gaasperdammerweg (0.532)
3. **Cluster 162** - Haarlemmerweg (0.514)

Results Summary:
- **0** High-risk predictions
- **94** Medium-risk predictions
- **76** Low-risk predictions

HARSH WEATHER SCENARIO

Weather Impact:
- Temperature: -10% decrease
- Precipitation: +30% increase
- Wind: +20% increase

Risk Changes:
- Average increase: +0.037 points
- Top score: 0.592 (Cluster 117)
- All hotspots: Elevated risk



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Key Findings Future Enhancements

Major Discoveries

- 66% of accidents in 170 specific clusters
- 95%accident increase (2017-2023)
- 6.7% risk increase in harsh weather
- Ringweg-West: 261 total accidents

Key Achievement

Transforming 30,000 accidents into actionable safety intelligence

Thank You - Questions Welcome

Actionable Intelligence:

Immediate Actions:
- Monitor top 10 predicted hotspots
- Enhance Ringweg-West infrastructure
- Deploy weather-based alerts
- Target safety campaigns

Future Enhancements:
- Real-time traffic integration
- Monthly prediction models
- Mobile cyclist app
- Expand to other Dutch cities

Research Extensions:
- Machine learning models
- Social factor integration
- Emergency response optimization

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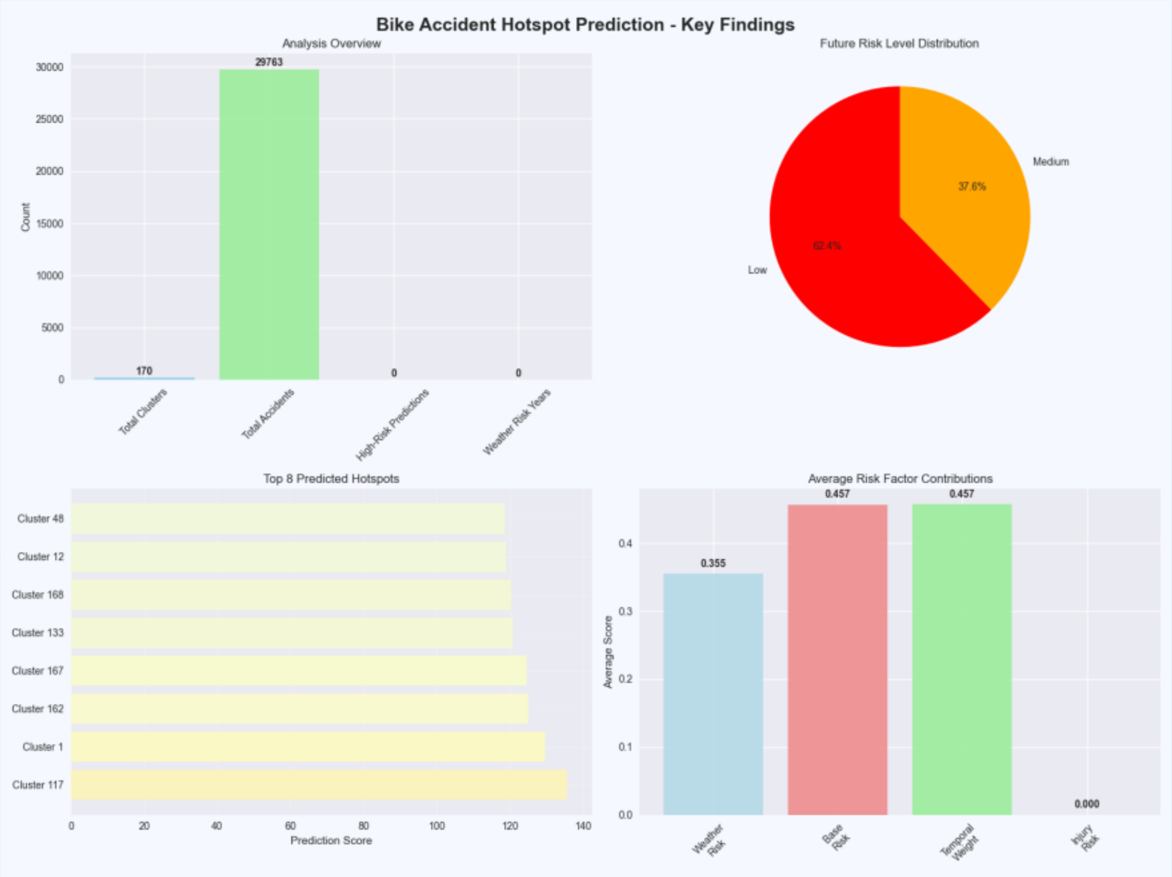

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